

Goldschmidt-based effective radii fit (lanthanides & actinides excluded) — Extended Report

Summary:

Data points used (total): 2898

Elements fitted (after exclusion): 45

Fitted scale factors: $f_{B2}=1.0402$, $f_{AB}=1.0251$, $f_{AA}=1.0410$, $f_{BB}=1.0000$

Per-structure metrics:

- B2: $n=1296$, $RMSE=0.0974 \text{ \AA}$, $MAE=0.0708 \text{ \AA}$, $bias=0.0027 \text{ \AA}$

- L12: $n=1602$, $RMSE=0.1288 \text{ \AA}$, $MAE=0.0883 \text{ \AA}$, $bias=-0.0018 \text{ \AA}$

B2 evaluation

count: 1296

rmse_A: 0.09740445167543167

mae_A: 0.07082051924925034

median_abs_A: 0.0586956469886073

mean_bias_A: 0.00273608717283382

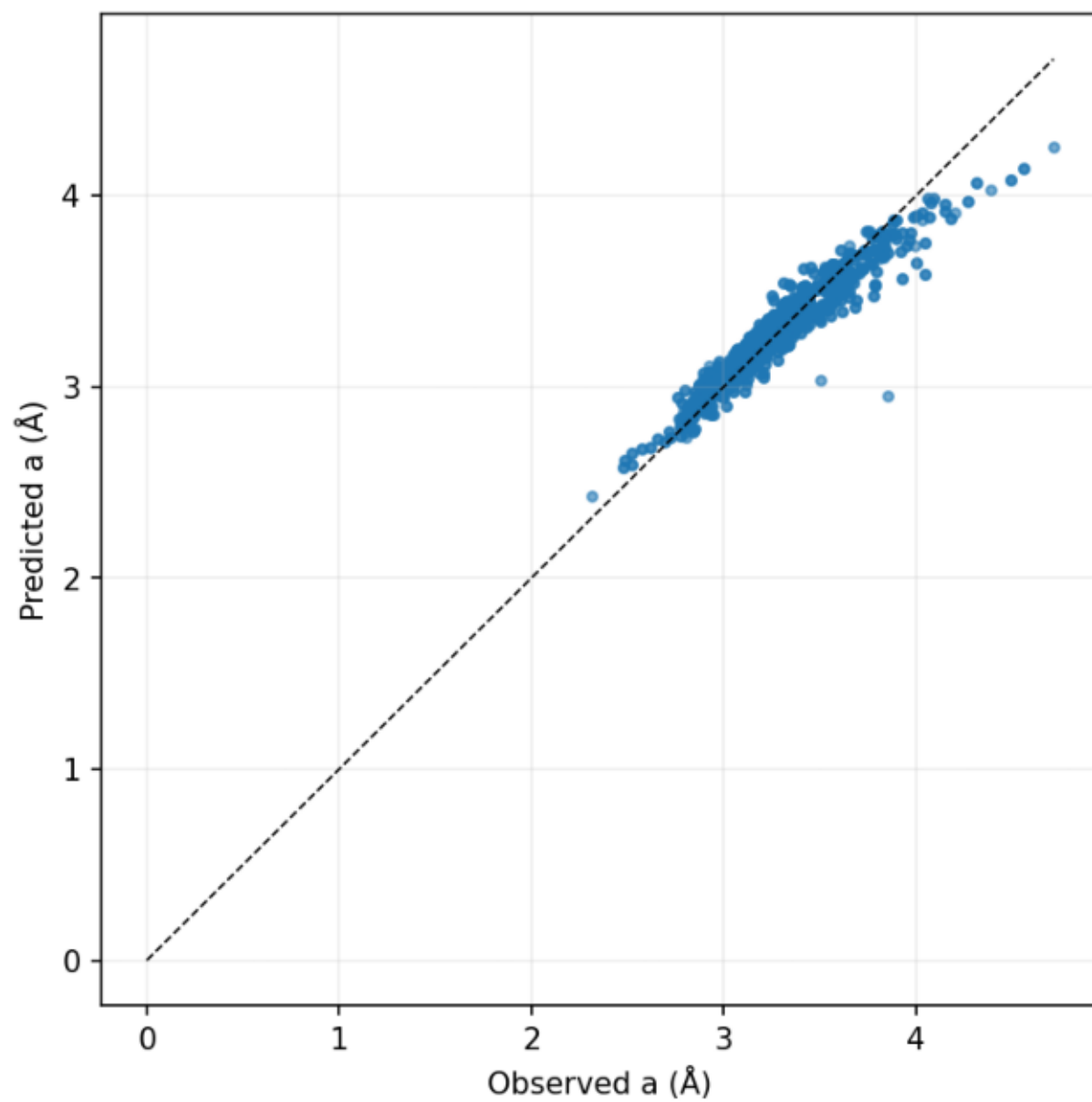
mean_rel_pct: 0.2394914259093276

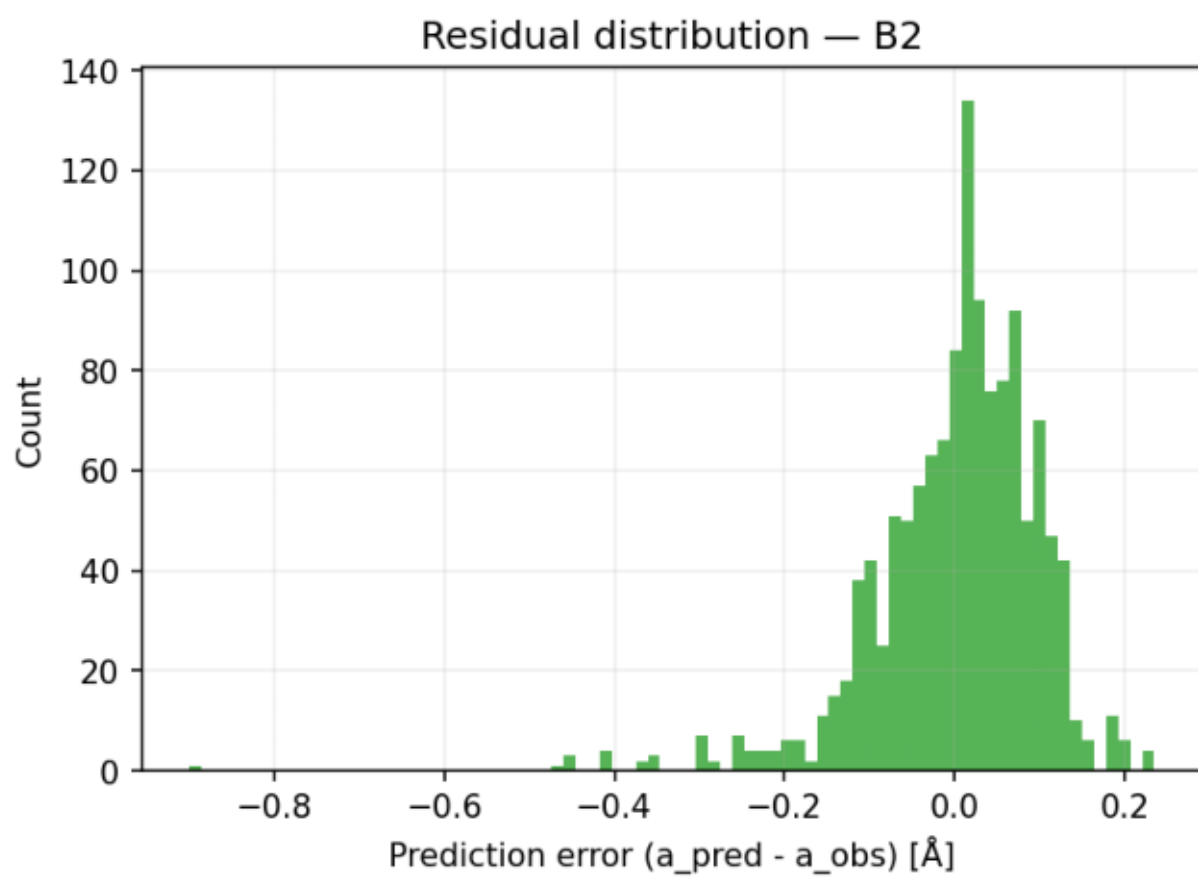
median_rel_pct: 0.5024804355714408

max_abs_A: 0.8992978106482665

p95_abs_A: 0.18164996767602526

Observed vs Predicted a — B2





Top outliers for B2 (top 15)

struct_type	elem_A	elem_B	a_obs	a_pred	element_order	abs_err	rel_err_pct
B2	Co	Mo	3.84973	2.950432	Co1Mo1	0.899298	-23.360023
B2	As	Fe	3.50486	3.037242	As1Fe1	0.467618	-13.341987
B2	Sr	Sr	4.71334	4.255944	Sr1Sr1	0.457396	-9.704279
B2	Cr	Sr	4.04606	3.590219	Cr1Sr1	0.455841	-11.266301
B2	Sr	Cr	4.04606	3.590219	Sr1Cr1	0.455841	-11.266301
B2	Ca	Sr	4.55508	4.144444	Ca1Sr1	0.410636	-9.014892
B2	Sr	Ca	4.55476	4.144444	Sr1Ca1	0.410316	-9.008499
B2	Sr	Na	4.49019	4.082654	Sr1Na1	0.407536	-9.076131
B2	Na	Sr	4.49005	4.082654	Na1Sr1	0.407396	-9.073296
B2	Sr	Mn	3.92574	3.564000	Sr1Mn1	0.361740	-9.214581
B2	Mn	Sr	3.92574	3.564000	Mn1Sr1	0.361740	-9.214581
B2	Sr	V	3.99640	3.644189	Sr1V1	0.352211	-8.813195
B2	V	Sr	3.99636	3.644189	V1Sr1	0.352171	-8.812283
B2	Ca	Ca	4.38383	4.032945	Ca1Ca1	0.350885	-8.004083
B2	Mg	Sr	4.17849	3.876229	Mg1Sr1	0.302261	-7.233728

L12 evaluation

count: 1602

rmse_A: 0.12884570998059322

mae_A: 0.08831429125354223

median_abs_A: 0.06624413959581199

mean_bias_A: -0.001811712681240958

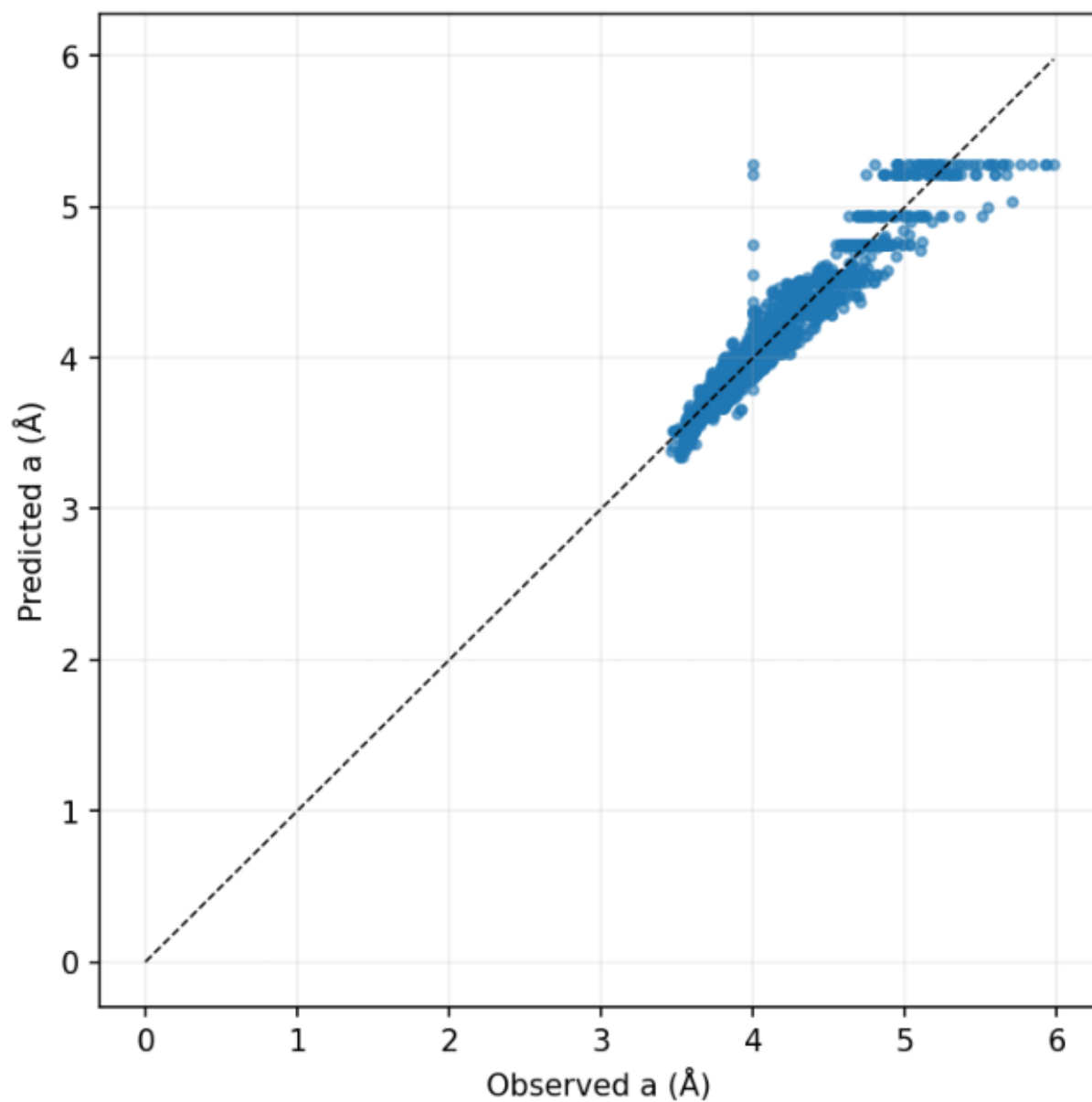
mean_rel_pct: -0.007124870174682608

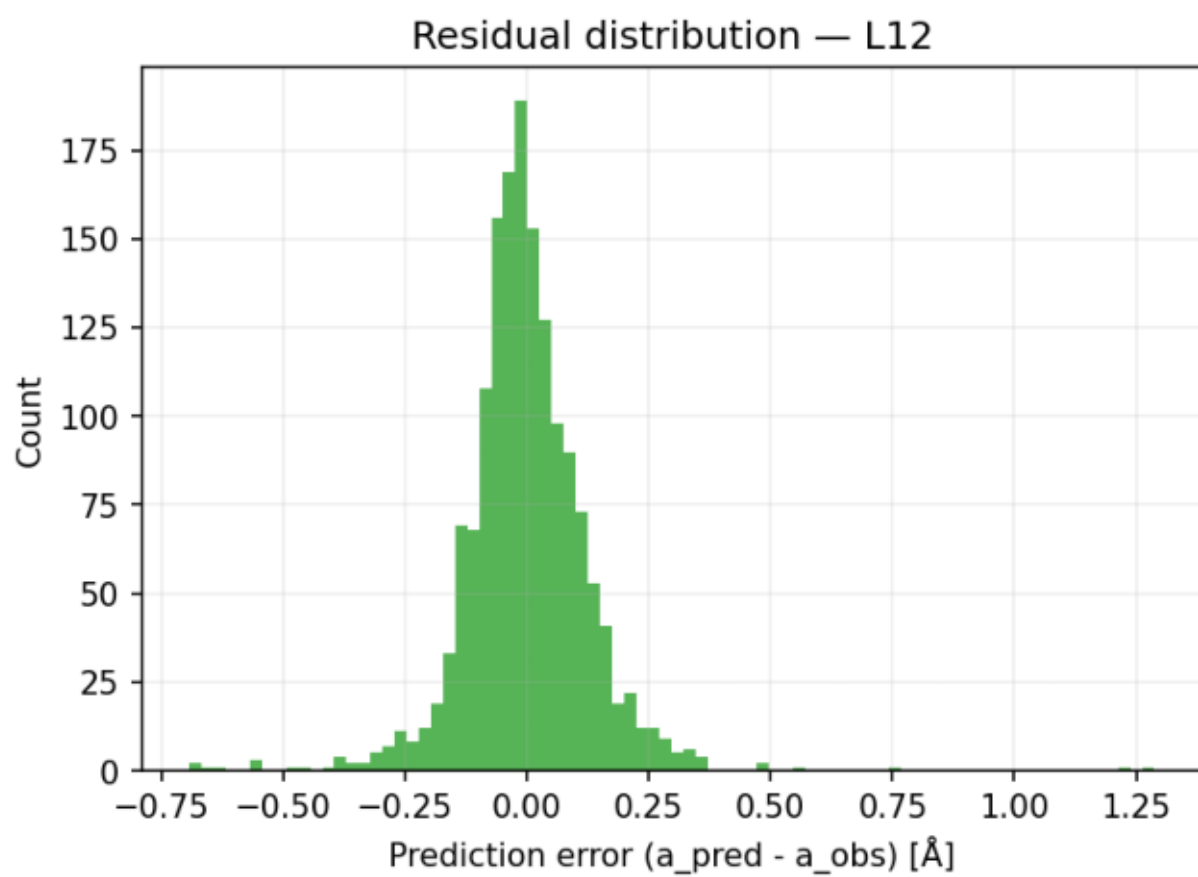
median_rel_pct: -0.2401858928625744

max_abs_A: 1.2872090919020698

p95_abs_A: 0.25303811248098024

Observed vs Predicted a — L12





Top outliers for L12 (top 15)

struct_type	elem_A	elem_B	a_obs	a_pred	element_order	abs_err	rel_err_pct
L12	Ba	Re	4.00000	5.287209	Re1Ba1Ba1Ba1	1.287209	32.180227
L12	Sr	Re	4.00000	5.216515	Re1Sr1Sr1Sr1	1.216515	30.412864
L12	Y	Re	4.00000	4.746694	Re1Y1Y1Y1	0.746694	18.667361
L12	Ba	Ba	5.97938	5.287209	Ba1Ba1Ba1Ba1	0.692171	-11.575965
L12	Ca	Ba	5.70938	5.036826	Ba1Ca1Ca1Ca1	0.672554	-11.779810
L12	Ba	Ca	5.93502	5.287209	Ca1Ba1Ba1Ba1	0.647811	-10.915059
L12	Ba	V	5.92480	5.287209	V1Ba1Ba1Ba1	0.637591	-10.761391
L12	Ca	Ca	5.50907	4.943184	Ca1Ca1Ca1Ca1	0.565886	-10.271903
L12	Sn	Re	4.00000	4.555658	Re1Sn1Sn1Sn1	0.555658	13.891456
L12	Ba	Sr	5.83661	5.287209	Sr1Ba1Ba1Ba1	0.549401	-9.413014
L12	Ca	Sr	5.54792	5.002020	Sr1Ca1Ca1Ca1	0.545900	-9.839720
L12	Ba	Cu	4.80015	5.287209	Cu1Ba1Ba1Ba1	0.487059	10.146747
L12	Ba	Mg	5.76509	5.287209	Mg1Ba1Ba1Ba1	0.477881	-8.289219
L12	Sr	Cu	4.74461	5.216515	Cu1Sr1Sr1Sr1	0.471905	9.946119
L12	Sr	Ba	5.66655	5.216515	Ba1Sr1Sr1Sr1	0.450035	-7.941965